

PRATHMESH PATIL

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❖ Summary: -

UAV Systems Engineer with 20+ projects in drone integration, autonomy, embedded systems, and field ops. Strong skills in Pixhawk, Betaflight, PID tuning, telemetry, and diagnostics. **DGCA Certified Remote Pilot. World Rank 3 (FPV Race)** at TechnoXian 9.0. Currently building defence-grade UAV Systems at Droneacharya Aerial Innovations Pvt. Ltd.

❖ Education: -

- **Marathwada Mitra Mandal's College of Engineering, Pune** 2022-2026
Bachelor of Engineering in Electronics & Telecommunication (CGPA: 7/10)
- **Late P.B.Jog Jr. College** 2020-2022
12th MSBSHE (Percentage: 54.67%)
- **Sri Sri Ravishankar Vidya Mandir, Pune** 2010-2020
10th CBSE (Percentage: 85%)

❖ Experience: -

- **Captain, RC Drone Club: Team Vajra (MMCOE, Pune): -**
Led UAV projects covering system integration, autonomous missions, PID tuning and competition flight ops.
- **Intern (R&D), Nasan Medical Electronics Pvt. Ltd.: -**
Developed embedded/IoT prototypes, hardware debugging, and secure communication modules for medical devices.
- **UAV Systems Intern (R&D), Droneacharya Aerial Innovations Ltd.: -**
Built and deployed 7", 10" & 15" multi-role drone platforms for the Indian Armed Forces (grenade-drop & 5 kg heavy-lift ~30 min), worked with development of a tethered drone for ~12 hrs continuous flight, and contributed to agri-drone operations and 3D-printed custom components.

❖ Projects: -

- **Autonomous Quadcopter (Delivery Application): -**
 - Waypoint navigation, intelligent failsafes, custom telemetry module, and payload stability optimisation.
- **Pixhawk Quadcopter – Payload Delivery Platform: -**
 - PID tuning, flight testing, mission execution, and gripping mechanism for precision payload delivery.
- **High-Speed FPV Drone (SpeedyBee F405V3): -**
 - Assembled and tuned racing UAV for stable high-speed maneuvers; achieved World Rank 3 at TechnoXian 9.0.

❖ Achievements: -

- **World Rank 3rd** - FPV Drone Race, TechnoXian 9.0 World Robotics Championship 2025.
- **World Rank 5th** - Drone Rescue, TechnoXian 9.0 World Robotics Championship 2025.
- **DGCA Certified Remote Pilot (Small Cat.)**
- **Best Paper Award - IEEE ICSFT 2026:** "Smart Renewable Energy Management System with Over-The-Air (OTA) Update"

❖ Technical Skills: -

- **UAV Systems:** Pixhawk, Ardupilot, Betaflight, Mission Planner, PID tuning, tethered UAVs, payload & grenade-drop systems
- **Flight Ops:** Autonomous missions, FPV piloting, flight testing, failsafes
- **Embedded:** C, C++, Python, Embedded C, STM32, ESP32, Raspberry Pi, RS485, MODBUS, UART/SPI/I2C
- **Tools & Hardware:** STM32CubeIDE, KiCad, Mender, Balena, GPS/IMU/ESC debugging